Novel Method for Monocular Vision Based Mobile Robot Localization

Li Maohai, Hong Bingrong
Harbin Institute of Technology
School of Computer Science & Technology
Harbin 150001, China
limaohai@163.com

Luo Ronghua
South China University of Technology
School of Computer Science & Engineering
Guangzhou 510641, China

Abstract

A robust environment map with 3D spatial natural landmarks that facilitates monocular vision based mobile robot for global localization is built. The highly distinctive multi-dimensional vector descriptors associated with the features extracted through Scale Invariant Feature Transform (SIFT) can be robustly matched despite changes in illumination, scale and viewpoint. The landmarks are 3D restructured with the matching image feature pairs obtained through the KD-Tree based nearest search approach. Novel RANSAC approach based on generic optimization for global localization is presented. Experiments on the robot Pioneer3 with monocular vision in our real indoor environment show that our method is of high precision.

1. Introduction

The acquisition of unknown environment models, navigation and localization belong to the essential capabilities of mobile robot. There are two types of localization: local and global. Local techniques aim at compensating for odometry errors. They require that the initial location of the robot be approximately known. Global techniques can localize a robot without any prior knowledge about its position, i.e., they can handle the kidnapped robot problem. Global localization techniques are more powerful than local ones and allow the robot to recover from serious positioning errors.

To achieve localization, there are different types of sensors such as sonar, laser range finders and vision. Sonar is fast and cheap but usually very crude, whereas lasers are active, accurate but heavy, expensive, slow. Vision systems are passive and lightweight, small, and cheap, but are fairly difficult to work with. Many early successful approaches [1] utilize artificial landmarks, such as bar-code reflectors, ultrasonic beacons, visual patterns, etc., and therefore do not function properly in beacon free environments. Therefore, vision-based approaches using stable natural landmarks in unmodified environments are highly desirable for a wide range of applications.

Thrun [2] developed the museum tour-guide robot MINERVA that employs EM to learn its map and Markov localization with camera mosaics of the ceiling. Hayet [3] extracted and recognized planar quadrangular landmarks from images. Kosecka [4] employed gradient orientation histograms to capture the essential object appearance information. Sim [5] learnt natural visual features and used principal component analysis (PCA) to match landmarks for pose estimation. Cobzas [6] used a panoramic image-based model for robot localization. Most of the existing vision-based localization algorithms are based on probabilistic methods and use sensor information of low feature specificity which introduced complex data association problems, and require the robot to move around for the probabilities to gradually converge toward a peak, whereas our approach uses high-level image features which are scale invariant and highly distinctive, thus greatly facilitating feature correspondence and the localization problem [7].

In this paper, we propose a monocular vision-based mapping algorithm by tracking the scale-invariant feature transform (SIFT) visual landmarks in unmodified environments [8]. 3D positions of the landmarks are obtained through 3D restructure of matching SIFT feature pairs, and a fast and efficient algorithm for matching features in a KD-Tree [9] is presented. Hence, in the built 3D map, novel RANSAC [10] approach based on generic optimization for global localization is presented and implemented on the robot Pioneer3 in our real indoor environment, and experiment results show that our method is of high precision.

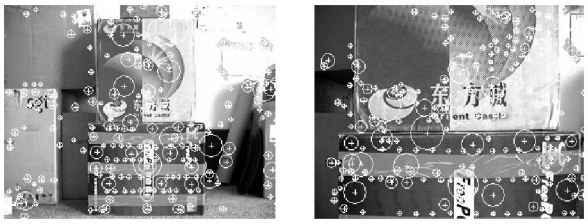
2. Environment model

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2.1. SIFT feature extraction

SIFT was proposed in [8] as a method of extracting and describing key-points, which are robustly invariant to common image transforms. The SIFT algorithm has four major stages: 1) Scale-space extrema detection. 2) Key-point localization. 3) Orientation assignment. 4) Key-point descriptor. An important aspect of the algorithm is that it generates a large number of highly distinctive features over a broad range of scales and locations. For a more detailed discussion see [8]. In this paper, we use the vectors with 128 elements as key-point descriptor. Fig. 1 shows an example of SIFT feature extraction for a cluttered and occluded image of size 320×240 pixels.



(a) (b)

Fig. 1. Typical extracted SIFT features with their locations represented by '+'. The radius of the circle represents their scales: the 320×240 pixel test image taken at (a) 1618mm; (b) 756mm; and (a) 278 key-points; (b) 267 key-points.

2.2. KD-Tree based feature matching

After SIFT feature extracting, SIFT feature matches are obtained in adjacent images through KD-tree based feature matching algorithm. Given a SIFT key-points set E, and a target key-point vector d, then a nearest neighbor of d, d' is defined as:

$$\forall d'' \in E, |d \leftrightarrow d'| \le |d \leftrightarrow d''|, \quad |d \leftrightarrow d'| = \sqrt{\sum_{i=1}^{k} (d_i \leftrightarrow d'_i)^2} \tag{1}$$

where d_i is the i-th component of d. The basic process for SIFT key-points matching based on nearest neighbor search algorithm in a KD-tree is as follows: A KD-tree is constructed using all key-points of the image I_t. For each key-point kp in the next image I_{t+1}, finding the two most nearest neighbors kp_1 and kp_2 based on nearest neighbor algorithm in a KD-tree. As proved in our experiment, if $|kp_1 \leftrightarrow kp|/|kp_2 \leftrightarrow kp|$ is bigger, then the matching quality between kp and kp_1 is much higher, otherwise the matching quality is lower. So we can use the following equation to judge the matching for two key-points:

$$|kp_1 \leftrightarrow kp| / |kp_2 \leftrightarrow kp| < \lambda \tag{2}$$

where λ is constant, and $0<\lambda<1$(in this paper λ is evaluated as 0.7). Fig. 2 shows an example of feature matching with different scale and direction, and we obtain 67 matches that the matched accurate rate is higher than 80%. Fig. 3 shows the key-point descriptor histograms of one matching pair in Fig. 2, which illustrates the robust matching algorithm.

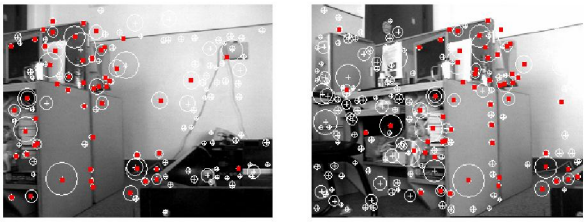


Fig. 2. The SIFT feature matches based on KD-tree, and the matches are represented by red ".".

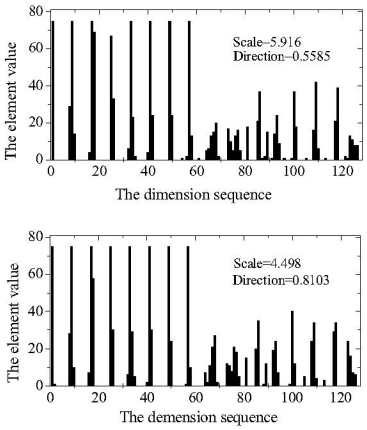


Fig. 3. The key-point descriptor histograms of the matching key-point at different scale and direction.

2.3. 3D restructure

As shown in Fig. 4.a, according to the epipolar constraint, all the entities P, O_1, O_2, p_1, p_2, e_1, e_2, b should be coplanar. Through epipolar constraint, the matches with large error are eliminated. Let f be the focus of the CCD camera. The relationship between a 3D point $P(X_w,Y_w,Z_w)$ and the image coordinates $p(u,v)$ where it is projected is given by the pinhole camera model [11]:

$$Z_c \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha_x & 0 & u_0 & 0 \\ 0 & \alpha_y & v_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \mathrm{R} & \mathrm{T} \\ 0^T & 1 \end{bmatrix} \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} = \mathrm{M} \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} \tag{3}$$

where motion model provides extrinsic camera rotations R and translations T for each image. Offline calibration yields the camera's intrinsic parameters α_x, α_y, u_0, v_0. For any pair of matching points $p_1(u_1,v_1,1)$

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and $p_2(u_2,v_2,1)$ corresponding to a 3D point $P(X_w,Y_w,Z_w)$, using the pinhole camera model:

$$\begin{aligned} z_{c1}[u_1\ v_1\ 1]^T &= \mathrm{M}[X_w\ Y_w\ Z_w\ 1]^T \\ z_{c2}[u_2\ v_2\ 1]^T &= \mathrm{M}[X_w\ Y_w\ Z_w\ 1]^T \end{aligned} \quad (4)$$

The solution of three unknown variants X_w, Y_w and Z_w can be obtained through the least square method.

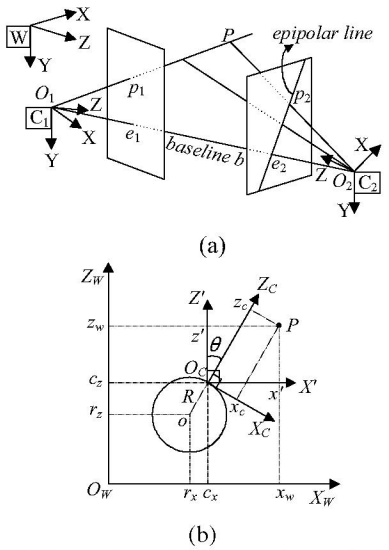


Fig. 4. (a) Two viewpoints geometry and the epipolar constraint; (b) The relation between word coordinates, camera coordinates and robot pose on 2D-plane.

For the robot moving on the approximate flat ground, we define coordinate axes X_w, Z_w, X_c and Z_c on the same coordinate plane as shown in Fig. 4.b. Let O_w and O_c be the origin of world coordinates and camera coordinates respectively, and the coordinates of O_c in the world be (c_x, c_z), the rotation angle θ, the world coordinates and camera coordinates of the spatial point P be (x_w, z_w) and (x_c, z_c) respectively. We set the assistant coordinates with axes X' and Z', then the coordinates (x', z') of P in assistant coordinates is equal to (x_w-c_x, z_w-c_z), and the relation between (x',z') and (x_c,z_c) is as following:

$$x_c = x'\cos\theta + z'\sin\theta \quad z_c = -x'\sin\theta + z'\cos\theta \quad (5)$$

For $y_c = y_w$, the final relation between (x_c, y_c, z_c) and (x_w, y_w, z_w) is represented as:

$$\begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} x_w - c_x \\ y_w \\ z_w - c_z \end{bmatrix} \quad (6)$$

We can obtain easily $c_x=r_x-R\sin\theta$ and $c_z=r_z-R\cos\theta$ from Fig. 4.b, where $R=z_{cr}$, then the above equation is transformed further to the follows about x_w, y_w and z_w:

$$\begin{aligned} &[\alpha_x\cos\theta-(u_0-u)\sin\theta]\cdot x_w+[\alpha_x\sin\theta+(u_0-u)\cos\theta]\cdot z_w \\ &=\alpha_x(r_x\cos\theta+r_z\sin\theta)-(u_0-u)(r_x\sin\theta-r_z\cos\theta-R) \\ &-(v_0-v)\sin\theta\cdot x_w+\alpha_y\cdot y_w+(v_0-v)\cos\theta\cdot z_w \\ &=-(v_0-v)(r_x\sin\theta-r_z\cos\theta-R) \end{aligned} \quad (7)$$

Then the landmarks' 3D spatial coordinates of corresponding SIFT feature matches are restructured.

3. Mobile robot localization

We only use the monocular vision to observe environment and build 3D map, where the landmarks is obtained through 3D restructure of matching SIFT feature pairs at different interval, so the landmark position accuracy is limit compared with using stereo vision. For estimating robustly the robot position, the image coordinates of SIFT feature should be sufficiently taken into consideration. Once the most likely relative position is estimated between the current view and model view, then the global robot position can be obtained further.

3.1. Relative pose estimation

Let $\{ f_1^{(i)} \sim f_1^{(j)},...,f_k^{(i)} \sim f_k^{(j)} \}$ be k key-points matches between current view V_i and model view V_j in map database, we can obtain $\tilde{I}_k^{(i)} F \tilde{I}_k^{(j)} = 0$ according to epipolar constraint, where $\tilde{I}_k = [u\ v\ 1]$ is the normal image coordinates of key-point f_k, and the fundamental matrix $F=K^{-T}EK^{-1}$, where K is the camera internal parameter matrix, E is the essential matrix. For taking into account the fact that the motion of the robot is restricted to a plane x-z, and robot rotate around the y-axis by an angle θ, E has the following sparse form:

$$E = [T]_\times R = \begin{bmatrix} 0 & -t_z & 0 \\ t_z & 0 & -t_x \\ 0 & t_x & 0 \end{bmatrix} \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix} \quad (8)$$

where $[T]_\times$ is the anti-symmetric matrix of translation $T=[t_x\ 0\ t_z]^T$, R is the corresponding rotation matrix. Obviously, once the essential matrix E has been recovered, the four different solutions for θ and $T=[t_x\ 0\ t_z]^T$ can be obtained through basic trigonometry directly from the parameter of the above equation. The physically correct solution is then obtained using the positive depth constraint. For robot moving on the plane, the fundamental matrix F can be recovered from the homogeneous epipolar constraint given a minimum of five matches $\{ f_1^{(i)} \sim f_1^{(j)},...,f_k^{(i)} \sim f_k^{(j)} \}_{k=5}$:

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$$\begin{cases} \tilde{I}_k^{(i)T} F \tilde{I}_k^{(j)} = (v_k^{(i)} f_{21} + f_{31}) u_k^{(j)} + (u_k^{(i)} f_{12} + f_{32}) v_k^{(j)} \\ \qquad\qquad + u_k^{(i)} f_{13} + v_k^{(i)} f_{23} + 1 = 0, \quad (k=1\sim5) \\ |F| = f_{12} f_{23} f_{31} + f_{13} f_{21} f_{32} - f_{12} f_{21} = 0 \end{cases} \tag{9}$$

where $(u_k^{(i)}, v_k^{(i)}, 1)$ and $(u_k^{(j)}, v_k^{(j)}, 1)$ is the k-th normal image coordinates of key-point $f_k^{(i)}$ and $f_k^{(j)}$ respectively. However, F obtained through five points is sensitive to noise, so we employ the RANSAC method to find the robust estimation of the fundamental matrix F. The detailed procedure is described as follows:

1) Tentative Matches: First, we create a list of tentative matches $\{ f_1^{(i)} \sim f_1^{(j)}, ..., f_k^{(i)} \sim f_k^{(j)} \}$ from the landmarks in the V_i to the landmarks in the V_j using the proposed KD-tree based feature matching method. For each matched landmarks, which should have the similar height in spatial y orientation.

2) Computing the Alignment: Next, we randomly select five tentative matches from the list and compute the alignment fundamental matrix F' from them according to equation (9). Five tentative matches are required at least in this case.

3) Seeking Support for F': Now we check all the tentative matches $\{f_1^{(i)} \sim f_1^{(j)}, ..., f_k^{(i)} \sim f_k^{(j)}\}$ which support this particular fundamental matrix F'. For each key-point match $\{f_k^{(i)} \sim f_k^{(j)}\}$, if $dist(I_k^{(i)}, F'I_k^{(j)}) \le \xi$ and $dist(I_k^{(j)}, F'I_k^{(i)}) \le \xi$ (ξ is the predetermined constant, in this paper, ξ=0.99), then match $\{f_k^{(i)} \sim f_k^{(j)}\}$ supports this particular matrix F', where $I_k^{(i)}$ and $I_k^{(j)}$ is the image coordinate of the feature $f_k^{(i)}$ and $f_k^{(j)}$ respectively, $F'I_k^{(i)}$ is the epipolar line on the image plane corresponding to $I_k^{(i)}$, and $F'I_k^{(j)}$ is the epipolar line on the image plane corresponding to $I_k^{(j)}$, $dist(\cdot)$ is the function of the distance between point and line.

4) Hypothesis With Most Support: The above three steps are repeated m times. The fundamental matrix with most support is our hypothesis. We then proceed with least-squares minimization for this hypothesis and obtain a better estimate for the fundamental matrix.

When the robot is facing a scene with very few SIFT landmarks, we can build a small sub-map of a *local* region from multiple frames instead and then align this sub-map to the database map. We employ an algorithm very similar to the above, when the robot wants to localize itself, it rotates a little bit, from -15° to 15° and builds a sub-map of this local region using information from multiple frames.

Once the essential matrix E has been recovered, the four different solutions for θ and T' can be obtained (using basic trigonometry) directly from the parametrization of the essential matrix (8), and $T'=cT$, where T is truly value and c is ratio constant.

3.2. Global pose estimation

Let $\{f_1^{(i)} \sim L_1^{(j)}, ..., f_k^{(i)} \sim L_k^{(j)}\}$ be k matches between key-points in current view V_i and 3D landmarks in map database. For these k 3D landmarks can be seen in different interval, let $\{I_1, ..., I_{k_i}\}_{t=t_i}$ be k_i image coordinates corresponding k 3D landmarks seen at time t_i, let $\{k_1, ..., k_n\}$ be the n maximal value for k_i at all times, then we can obtain $\{\theta_1, ..., \theta_n\}$ and $\{R_1, ..., R_n\}$ through relative position estimation. According to the projection equation, we can get the follows:

$$z_c^{(j)} \begin{bmatrix} u_i^{(j)} & v_i^{(j)} & 1 \end{bmatrix}^T = C \begin{bmatrix} R_i & T_i \end{bmatrix} \begin{bmatrix} \mathrm{x}_i^{(j)} & \mathrm{y}_i^{(j)} & \mathrm{z}_i^{(j)} & 1 \end{bmatrix}^T \tag{10}$$

where $(u_i^{(j)}, v_i^{(j)}, 1)^T$ and $(\mathrm{x}_i^{(j)}, \mathrm{y}_i^{(j)}, \mathrm{z}_i^{(j)}, 1)^T$ is the homogeneous image coordinates and 3D spatial coordinates of landmark $L_i^{(j)}$. For the robot moves around on the x-z plane, the translation $T_i=[x_i, 0, z_i]^T$ only contains the two unknown parameters x_i and z_i, then the above equation can be transformed as:

$$\begin{aligned} z_i &= ((m_{31}^{(i)} \mathrm{x}_i^{(j)} + m_{33}^{(i)} \mathrm{z}_i^{(j)}) v_i^{(j)} - a_y \mathrm{y}_i^{(j)}) \big/ (v_0^{(j)} - v_i^{(j)}) \\ x_i &= \left(\begin{array}{l} ((m_{31}^{(i)} \mathrm{x}_i^{(j)} + m_{33}^{(i)} \mathrm{z}_i^{(j)}) u_i^{(j)} - m_{11}^{(i)} \mathrm{x}_i^{(j)} \\ -m_{13}^{(i)} \mathrm{z}_i^{(j)} - (u_0^{(j)} - u_i^{(j)}) z_i \end{array} \right) \Big/ a_x \end{aligned} \tag{11}$$

where a_x, a_y is the intrinsic parameter of CCD camera, and $m_{lk}^{(i)}$ ($l, k \in \{1, 2, 3\}$) is the l-th row and the k-th column of projection matrix M_i, then each landmark can compute one estimated robot pose, we can obtain k_i pose estimates from k_i landmarks. For obtaining the robust solution of z_i and x_i, we select randomly two from k_i landmarks to find the solution through the least square method, which repeats H times, then we get nH estimated pose in total. The target for global localization is to select the best one from these nH pose, which is implemented as shown in Algorithm 1.

The crossover and mutation operator in generic optimization are defined as following:

Crossover: select two parent pose $(s_t^{(a)}, w_t^{(a)})$ and $(s_t^{(b)}, w_t^{(b)})$ randomly from nH pose, the crossover operator mates them by the following equation to generate two children $(s_t^{(c)}, w_t^{(c)})$ and $(s_t^{(d)}, w_t^{(d)})$:

$$\begin{cases} s_t^{(c)} = \kappa s_t^{(a)} + (1-\kappa) s_t^{(b)} + \tau, & w_t^{(c)} = p(z_t \mid s_t^{(c)}) \\ s_t^{(d)} = \kappa s_t^{(b)} + (1-\kappa) s_t^{(a)} + \tau, & w_t^{(d)} = p(z_t \mid s_t^{(d)}) \end{cases} \tag{12}$$

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where $\kappa \sim U[0,1]$, $\tau \sim N(0,\Sigma)$, and $U[0,1]$ represents uniform distribution and $N(0,\Sigma)$ the normal distribution. Then, replace the parents $\{s_t^{(a)}, s_t^{(b)}\}$ by theirs children $\{s_t^{(c)}, s_t^{(d)}\}$ according to the following criterion: The child $s_t^{(c)}$ would be accepted if $p(z_t|s_t^{(c)}) > \max(p(z_t|s_t^{(a)}), p(z_t|s_t^{(b)}))$ value, else would be accepted with probability $\frac{p(z_t|s_t^{(c)})}{\max(p(z_t|s_t^{(a)}), p(z_t|s_t^{(b)}))}$. In the similar form is accepted or rejected the child $s_t^{(d)}$.

Algorithm 1 RANSAC approach based on generic optimization.

Input: Image frame and map database;
Output: the global robot localization;
1. Feature extraction from input image with SIFT;
2. KD-tree based feature matching between extracted feature with map;
3. Estimate $\{\theta_1, ..., \theta_n\}$ through relative pose estimation approach;
4. Obtaining nH pose as initial population according to Eq.12;
5. Population evolution;
6. The final result is voted through RANSAC approach.

Mutation: select one parent $(s_t^{(p)}, w_t^{(p)})$, the mutation operator on it is defined as following:

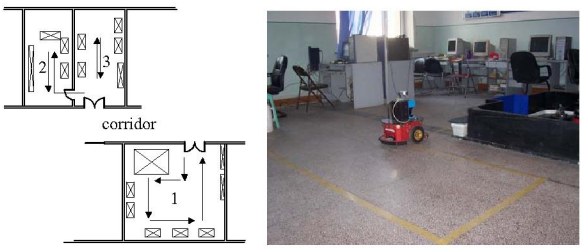
$$s_t^{(c)} = s_t^{(p)} + \sigma, \quad w_t^{(c)} = p(z_t \mid s_t^{(c)}), \quad \sigma \sim N(0,\Sigma) \quad (13)$$

Then $s_t^{(c)}$ is accepted if $p(z_t|s_t^{(c)}) > p(z_t|s_t^{(p)})$, else is accepted with probability $p(z_t|s_t^{(c)})/p(z_t|s_t^{(p)})$. Let N_L landmarks can be seen when robot locates at individual pose $s_t^{(c)}$, whose fitness value $w_t^{(c)}$ is defined as:

$$w^{(c)} = p(z \mid s^{(c)}) = \frac{N_L}{N_{\max}} - \prod_{j=1}^{N_L} p(I^{(j)} \mid L^{(j)}, s^{(c)}) \quad (14)$$

4. Experimental results

The experiments are performed on a Pioneer3 robot incorporating an 800 MHz Intel Pentium processor. Motor control is performed on the on-board computer, while a 3.0 GHz PC connected to the robot by a wireless link provides the main processing power for vision processing and the SLAM software. A single CCD camera mounted at the front of the robot is used for detecting the landmarks. The test environment is a medium size indoor space composed of three rooms and a horizontal corridor is shown in Fig.5.a, where the room 1 is shown in Fig.5.b.



(a) (b)

Fig. 5. (a) experiment model, (b) the real field for room 1.

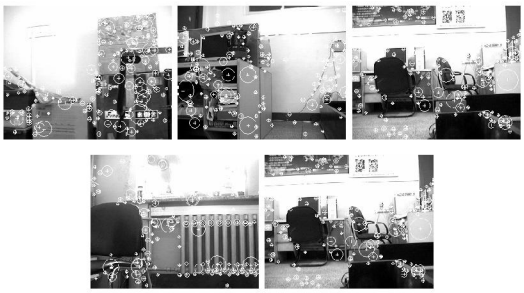
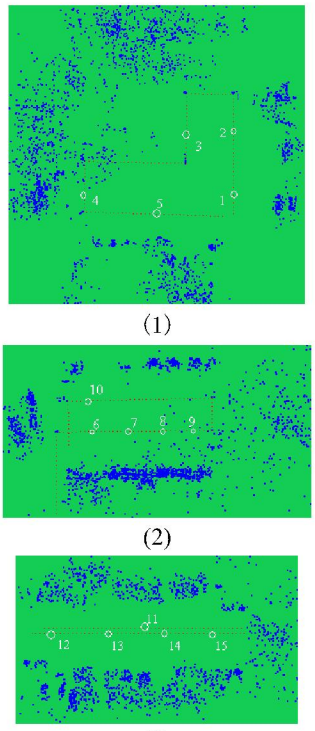


Fig. 6. Partial frames of an image sequence.



(1)

(2)

(3)

Fig. 7. The global localization experiments.

Partial captured image frames of size 320×240 are shown in Fig. 6. Fig. 7 shows the bird's eye view of the 3D feature map for three rooms. We implement the robot global localization by placing it at 5 positions in each room, and we test 20 times at each position. Fig. 8 shows the Gaussian distribution and Histogram for

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robot position error of global localization. As shown in Fig. 8, the average Euclidean translation error is less than 10cm, and the average rotation error is less than 4°. These errors are sufficient for our navigation requirement, which indicate further superior performance of presented method.

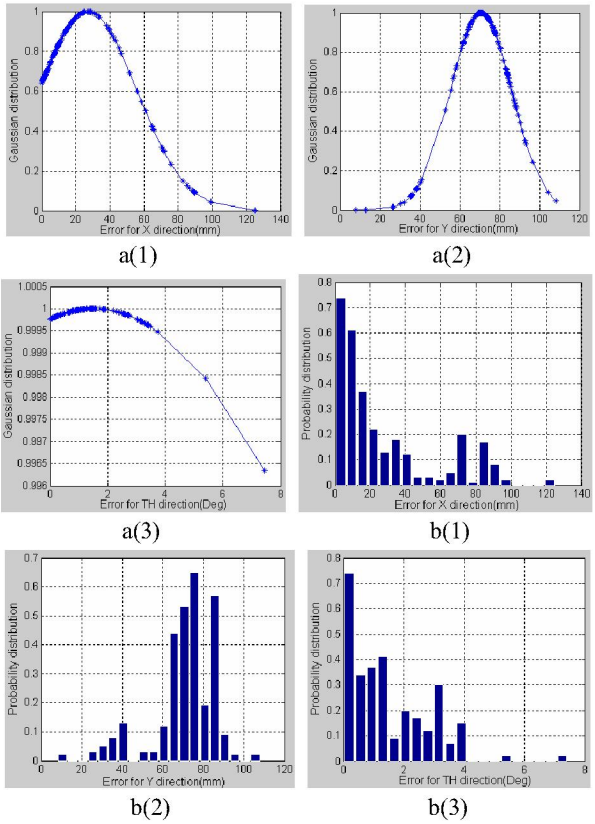


Fig. 8. (a) error Gaussian distribution; (b) error Histogram.

5. Conclusion

A robust environment map with 3D spatial natural landmarks that facilitates monocular vision based mobile robot for global localization is built. The highly distinctive multi-dimensional vector descriptors associated with the features extracted through SIFT can be robustly matched despite changes in illumination, scale and viewpoint. The landmarks are 3D restructured with the matching image feature pairs obtained through the KD-Tree based nearest search approach. Novel RANSAC approach based on generic optimization for global localization is presented. Experiments on the robot Pioneer3 with monocular vision in our real indoor environment show that our method is of high precision.

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